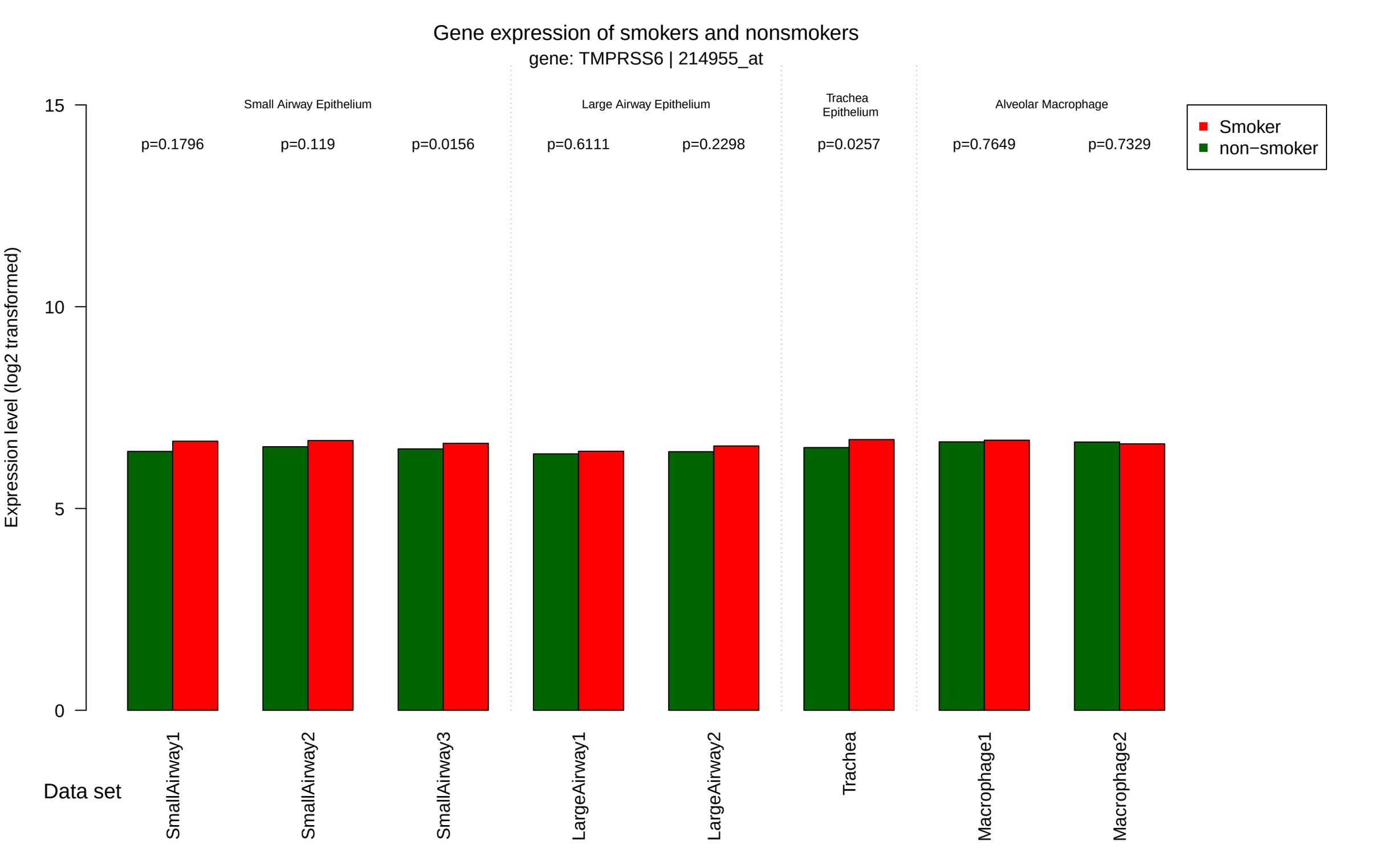
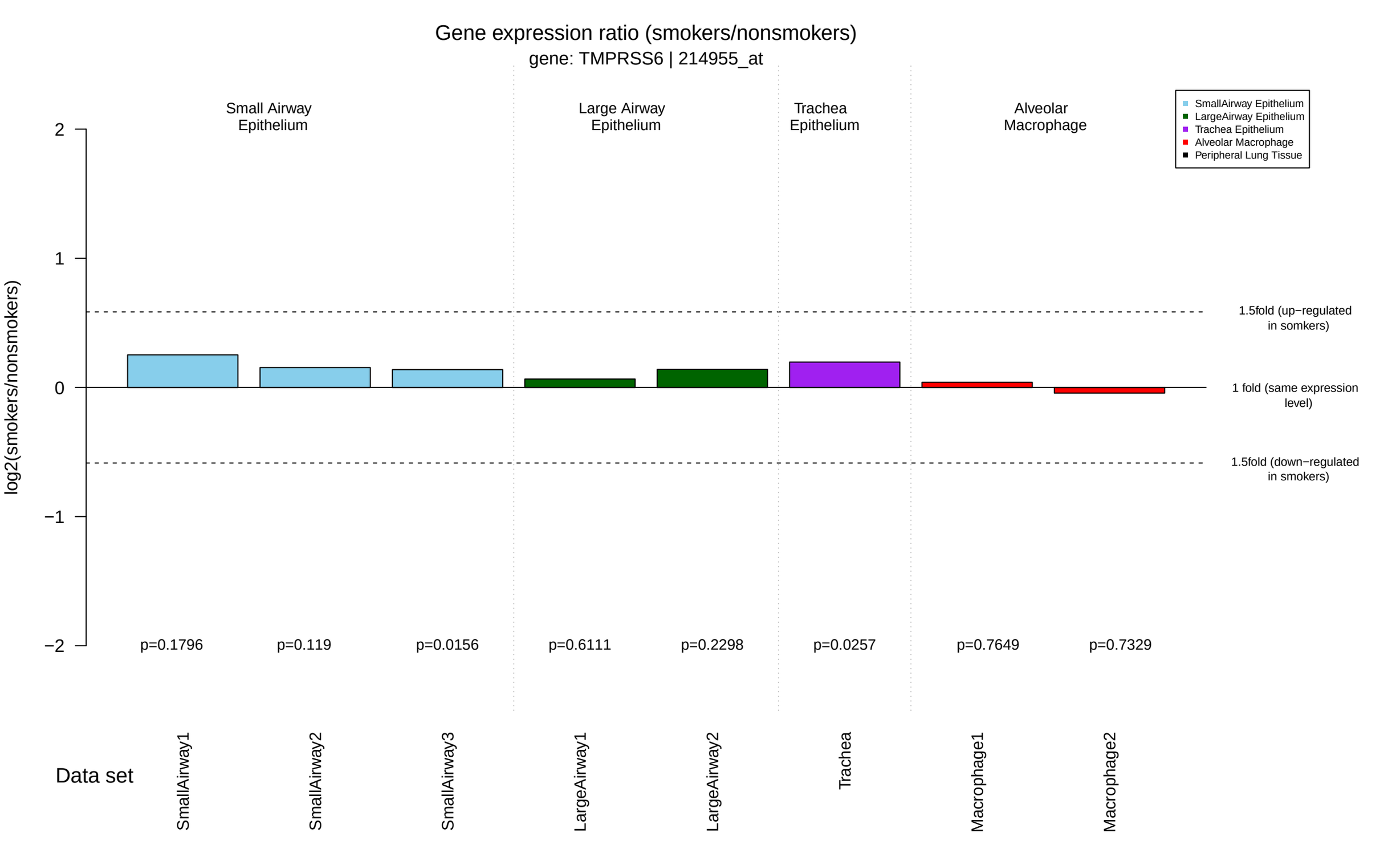
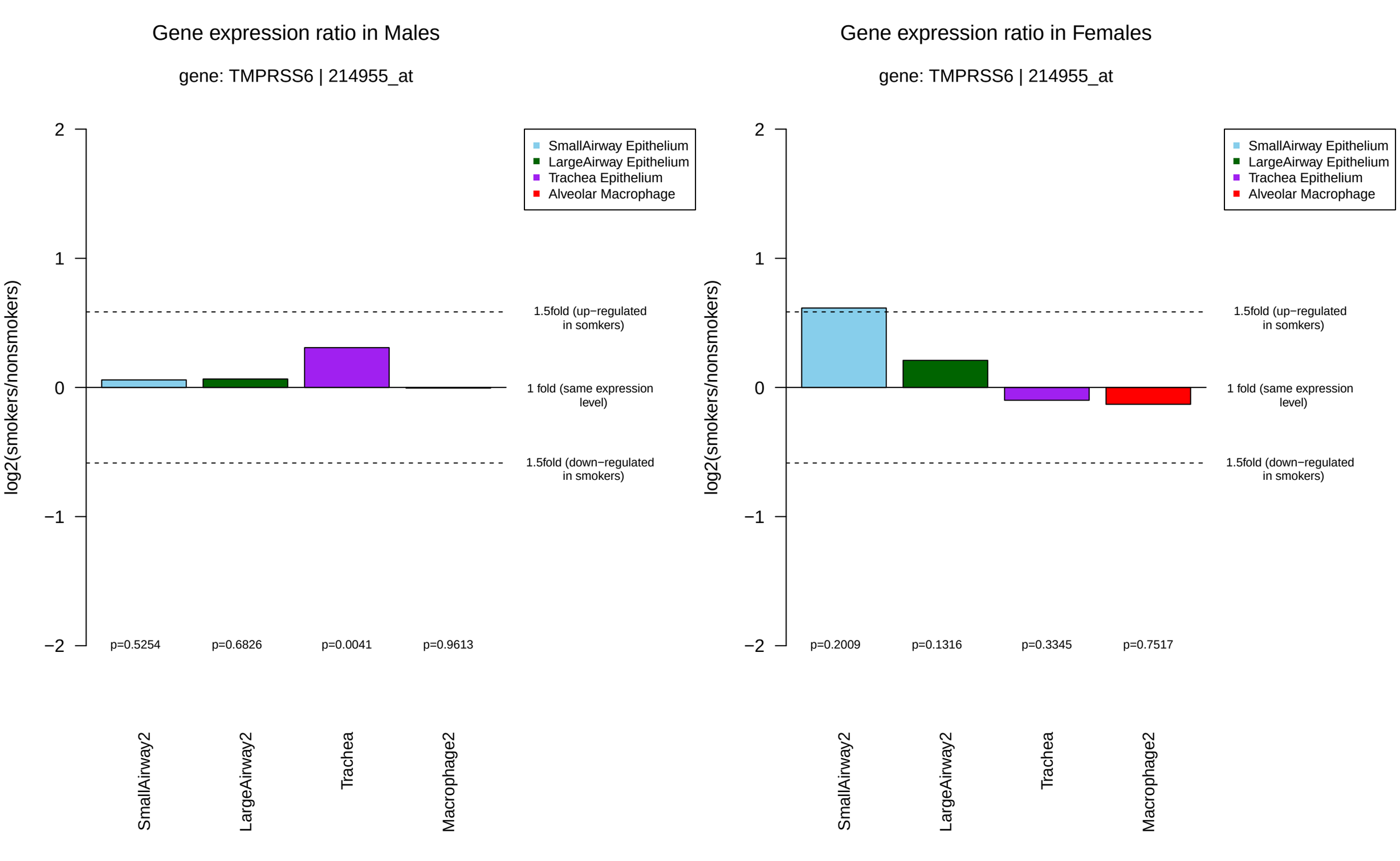
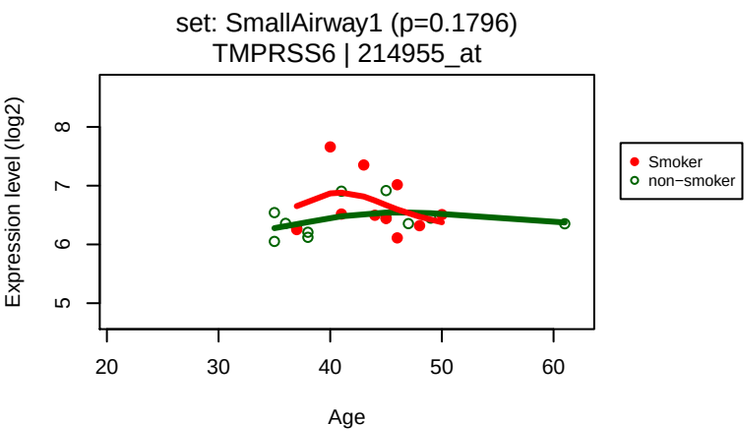
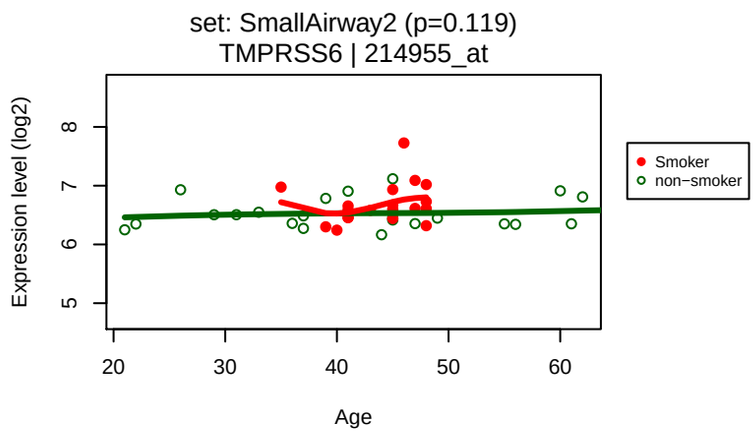
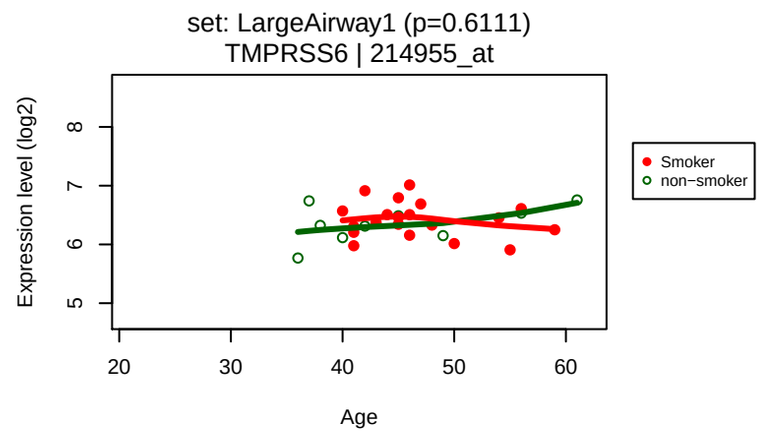

Gene expression of smokers and nonsmokers
gene: TMPRSS6 | 214955_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1796
p=0.119
p=0.0156
p=0.6111
p=0.2298
p=0.0257
p=0.7649
p=0.7329
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

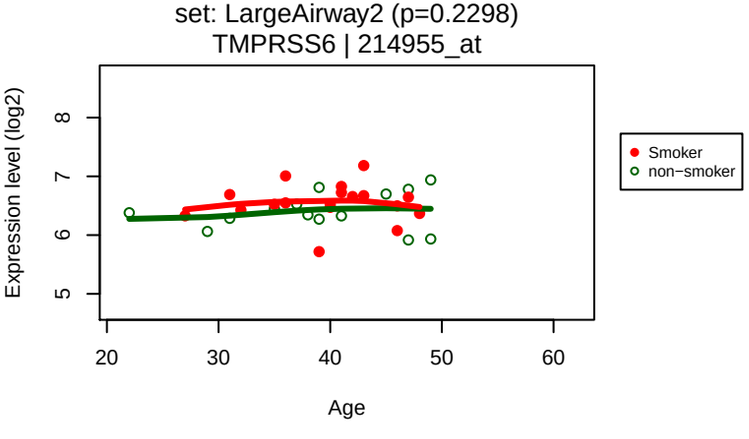

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS6 | 214955_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1796
p=0.119
p=0.0156
p=0.6111
p=0.2298
p=0.0257
p=0.7649
p=0.7329
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

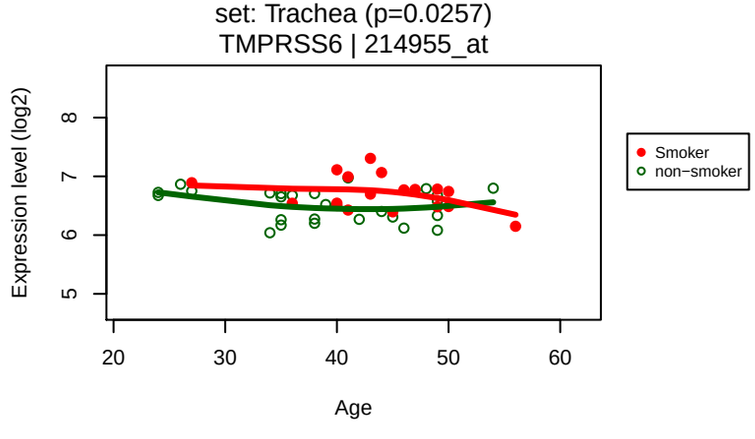

Gene expression ratio in Males
gene: TMPRSS6 | 214955_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5254
p=0.6826
p=0.0041
p=0.9613
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS6 | 214955_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2009
p=0.1316
p=0.3345
p=0.7517
SmallAirway2
LargeAirway2
Trachea
Macrophage2

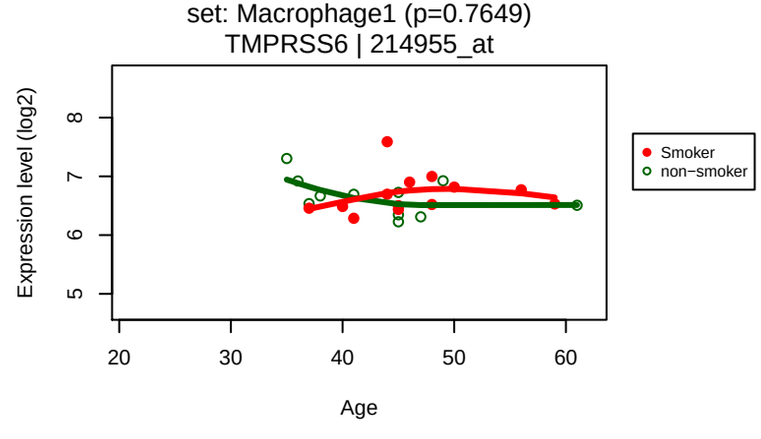

set: SmallAirway1 (p=0.1796)
TMPRSS6 | 214955_at
Expression level (log2)
Age
Smoker
non−smoker

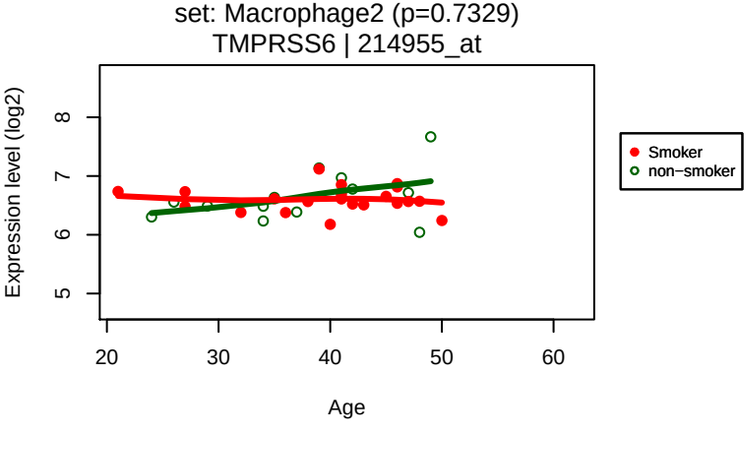

set: SmallAirway2 (p=0.119)
TMPRSS6 | 214955_at
Expression level (log2)
Age
Smoker
non−smoker

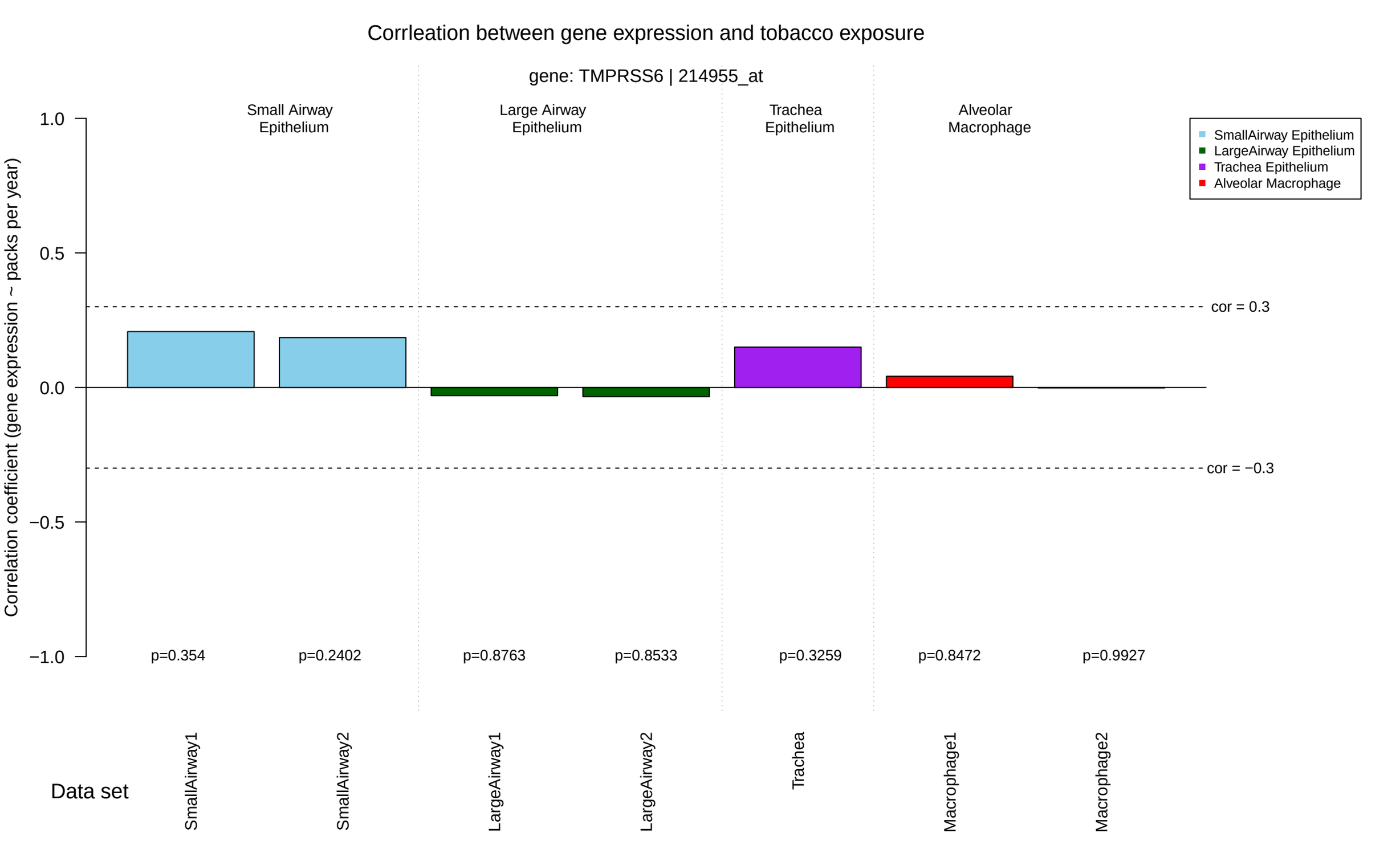

set: LargeAirway1 (p=0.6111)
TMPRSS6 | 214955_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2298)
TMPRSS6 | 214955_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0257)
TMPRSS6 | 214955_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7649)
TMPRSS6 | 214955_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7329)
TMPRSS6 | 214955_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS6 | 214955_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.354
p=0.2402
p=0.8763
p=0.8533
p=0.3259
p=0.8472
p=0.9927
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2